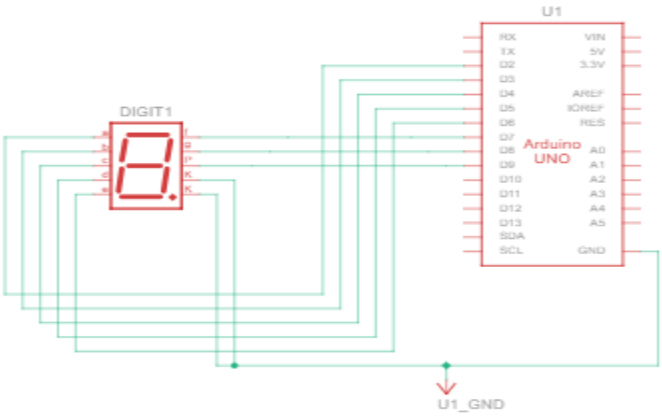
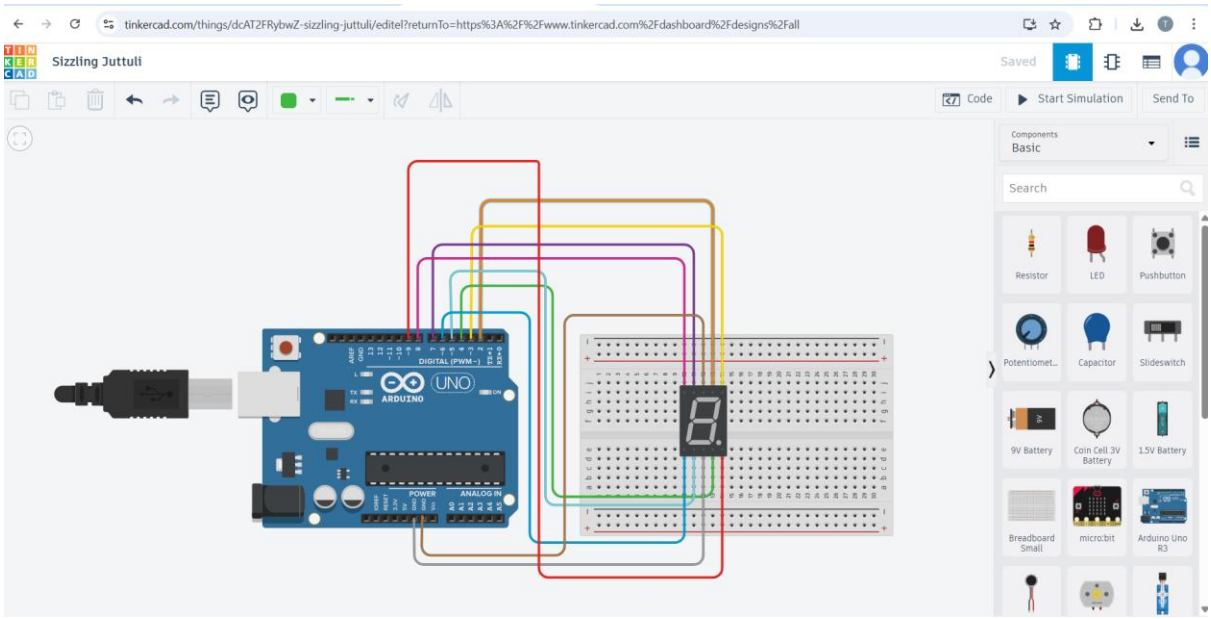
MAJOR PROJECT

Design an Arduino-based embedded program using a single-digit seven-segment display. The program should sequentially turn ON each segment one by one, starting with all segments OFF. Implement a fixed, predefined delay between each segment activation. Once all seven segments are ON, reverse the pattern by turning OFF each segment sequentially until all segments are OFF again. The sequence should then repeat continuously.

Schematic Representation



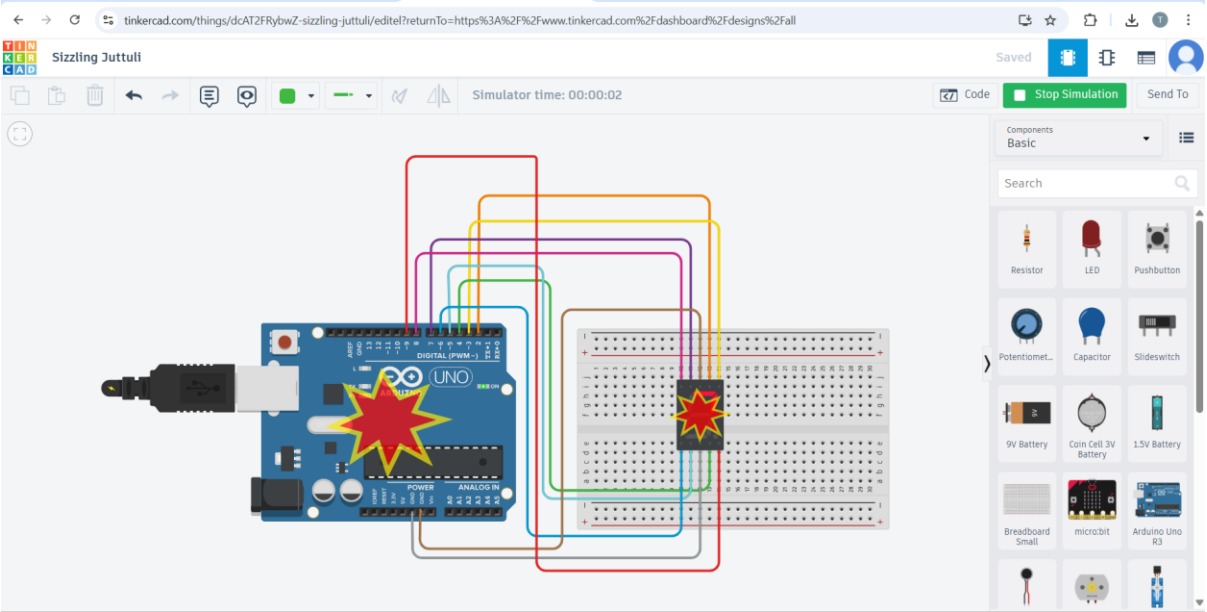
Circuit Diagram



Code for Execution

```
int pins[] = {2, 3, 4, 5, 6, 7, 8};
void setup() {
  for (int i = 0; i < 7; i++) {
    pinMode(pins[i], OUTPUT);
    digitalWrite(pins[i], LOW);
  }
}
void loop() {
  for (int i = 0; i < 7; i++) {
    digitalWrite(pins[i], HIGH);
    if (i >= 5) {
      delay(200);
    } else {
      delay(1000);
    }
  }
  for (int i = 6; i >= 0; i--) {
    digitalWrite(pins[i], LOW);
    if (i >= 5) {
      delay(200);
    } else {
      delay(1000);
    }
  }
}
```

Output



Link for output: https://videotourl.com/videos/1775221546566-7f1b6d13-dbc1-485f-80f3-deea03abd732.mp4